

Buffer Capacity Utilization: Boosting Throughput in Rivian's General Assembly

ISD (MFG) 503 Project Report

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12/11/2023

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Executive Summary

The focal point of my extensive summer internship at Rivian involved a multifaceted project aimed at the strategic enhancement of General Assembly operations within the organization. Initially, I was entrusted with the responsibility of streamlining production and mitigating downtime on four specific assembly lines: Trim2, Trim3, Final1, and Final2. However, the project's scope underwent a substantial transformation as part of Rivian's strategic drive to escalate production capacity. The new imperative was to ensure a daily output of 300 vehicles from the Final2 assembly line, regardless of potential bottlenecks in the process.

To effectively achieve this ambitious production target, I embarked on a comprehensive review of the entirety of the General Assembly layout. This review encompassed the integration of buffer mechanisms and a meticulous recalibration of cycle times, aligning them with the stringent requirement of 240 seconds for the Final2 assembly line. This recalibration initiated a cascading effect, necessitating the progressive reduction of cycle times for preceding assembly lines.

Subsequently, my professional focus transitioned to the pivotal task of station rebalancing, ensuring their alignment with the newly defined takt time. This process entailed close collaboration with cross-functional teams comprising material flow engineers, production control specialists, operations professionals, maintenance experts, material handling specialists, process engineers, and manufacturing engineers.

1.0 Introduction and Current State

This report outlines a strategic initiative conducted during a summer internship at Rivian to optimize General Assembly operations and boost throughput. The focus shifted towards achieving a daily output of 300 vehicles from the Final2 assembly line, necessitating a recalibration of cycle times and the integration of buffer mechanisms. The report delves into a systematic review of the General Assembly layout, emphasizing the utilization of buffer capacity to overcome bottlenecks and meet production targets. Collaboration with cross-functional teams was pivotal in executing this strategic overhaul, presenting an opportunity to enhance overall efficiency in Rivian's General Assembly.

1.1 Overview of the Project

In this strategic initiative at Rivian, General Assembly throughput optimization took precedence. The emphasis involved recalibrating cycle times and leveraging buffer capacity to achieve a daily output goal of 300 vehicles from the Final2 assembly line. The systematic review of the General Assembly layout, coupled with precision buffer utilization, aimed to address bottlenecks. Collaboration with cross-functional teams, including material flow engineers and manufacturing specialists, played a key role in this technical enhancement for increased efficiency.

Stakeholders: Understanding the project's stakeholders is essential:

- **Management and Leadership Team:** This group comprises the senior executives, directors, and managers within the General Assembly and Manufacturing Operations. They are key stakeholders as they provide strategic direction, make critical decisions, and ultimately approve project proposals and implementations.
- **General Assembly Personnel:** This category includes the assembly line workers, station operators, and other frontline staff directly impacted by changes in station layouts, cycle times, and processes. Their feedback and cooperation are essential for the successful implementation of the project.
- **Cross-Functional Teams:** These teams consist of various specialists, including material flow engineers, production control experts, operations professionals, maintenance and controls teams, material

handling specialists, process engineers, and manufacturing engineers. They are directly involved in implementing station rebalances and adjusting processes, making them crucial stakeholders.

- **Industrial Engineering Team:** As the owners of cycle times and process data, the Industrial Engineering team plays a significant role in maintaining the integrity and accuracy of the project's data. They are responsible for updating cycle time databases and ensuring the long-term sustainability of changes.
- **Internship Program Coordinators:** Within the context of an academic internship, my university's program coordinators are stakeholders as they oversee the learning objectives and outcomes of the internship experience.
- **Customers and Shareholders:** While not directly involved in the day-to-day project activities, customers and shareholders have a stake in the company's overall success, which is influenced by increased production capacity and revenue growth.
- **Regulatory and Compliance Authorities:** Depending on the nature of the automotive industry and manufacturing processes, regulatory and compliance authorities may have an interest in ensuring that changes made comply with industry standards and regulations.
- **Suppliers and Vendors:** Suppliers and vendors providing parts and materials to Rivian's General Assembly may be indirectly impacted by production changes and must adapt their supply chain processes accordingly.
- **Other Internal and External Parties:** Depending on the project's specific details, other internal and external stakeholders, such as unions, local communities, and governmental bodies, may be affected or have an interest in the project.

What is the Problem: The problem at hand is the optimization of General Assembly operations at Rivian, with a specific focus on increasing the daily production output of 300 vehicles from the Final2 assembly line. While this may appear as a numerical target, it encompasses a complex web of challenges.

Why It's Important:

1. **Production Efficiency:** Meeting the target of 300 vehicles per day from the Final2 assembly line is a pivotal aspect of production efficiency. It ensures that Rivian's production processes are finely tuned and capable of delivering their products at the desired rate. This is vital in meeting customer demand and staying competitive in the automotive market.
2. **Revenue Growth:** Increasing the production capacity to 300 vehicles per day translates into significant revenue growth. For Rivian, achieving this target results in an estimated annual revenue increase of approximately \$300 million. This substantial growth is paramount for the financial health and sustainability of the company, which ultimately benefits stakeholders, including shareholders, employees, and the company itself.
3. **Optimized Resource Utilization:** The optimization of General Assembly operations is essential for the efficient use of resources, including labor, equipment, and time. By reducing cycle times and eliminating bottlenecks, the organization can maximize the utilization of these resources, thereby improving cost-effectiveness and competitiveness.
4. **Customer Satisfaction:** Ensuring the timely production of vehicles is directly linked to customer satisfaction. Meeting production targets allows Rivian to deliver vehicles to customers as promised, building trust and a reputation in a highly competitive industry.
5. **Competitive Advantage:** In the automotive sector, where competition is fierce, the ability to produce more vehicles efficiently is a significant competitive advantage. It can lead to market leadership, attracting more customers, and outperforming rival companies.
6. **Strategic Goals:** Achieving the daily output target aligns with Rivian's strategic goals, which might include expansion, market share growth, or broader economic and environmental sustainability initiatives. It's a critical milestone in the organization's strategic roadmap.

1.2 Review of Current State Design / Literature Review

The current state of the project in optimizing General Assembly operations at Rivian can be summarized as follows:

1. Baseline Assessment: At the outset of the project, it was observed that the entire plant, including the General Assembly as a whole, was operating at a constant speed of 240 seconds per cycle, producing 15 vehicles per hour (JPH).

2. New Target and Objective: The project's objective shifted dramatically as the company set an ambitious goal to produce 300 vehicles daily from the Final2 assembly line, even if bottlenecks occur at other stages. This required a significant reconfiguration of production processes.

3. Layout and Buffer Integration: A detailed review of the General Assembly layout was conducted to accommodate this change. Buffers were introduced strategically to manage variations and bottlenecks.

4. Cycle Time Recalibration: In response to the new daily target, cycle times were recalibrated. The Final2 assembly line was set to 240 seconds per cycle, which necessitated reducing cycle times for all preceding assembly lines, creating a cascading effect.

5. Cross-Functional Collaboration: A critical aspect of the project involves close collaboration with various teams, including material flow engineers, production control, operations, maintenance, material handling, process engineers, and manufacturing engineers, to ensure successful implementation of the changes.

6. Trials and Approvals: The project is currently in the trial phase for the rebalanced stations. Once the trials are successfully completed, the 'before and after' changes will be presented to the leadership team for approval.

7. Implementation and Monitoring: The project is scheduled for implementation during a planned plant shutdown in mid-November. Following implementation, a systematic approach using the DMAIC methodology will be employed to control and analyze the changes.

8. Data Management: The Industrial Engineering team will play a pivotal role in updating the cycle time database in Delmia software, as they are responsible for maintaining and controlling cycle time information.

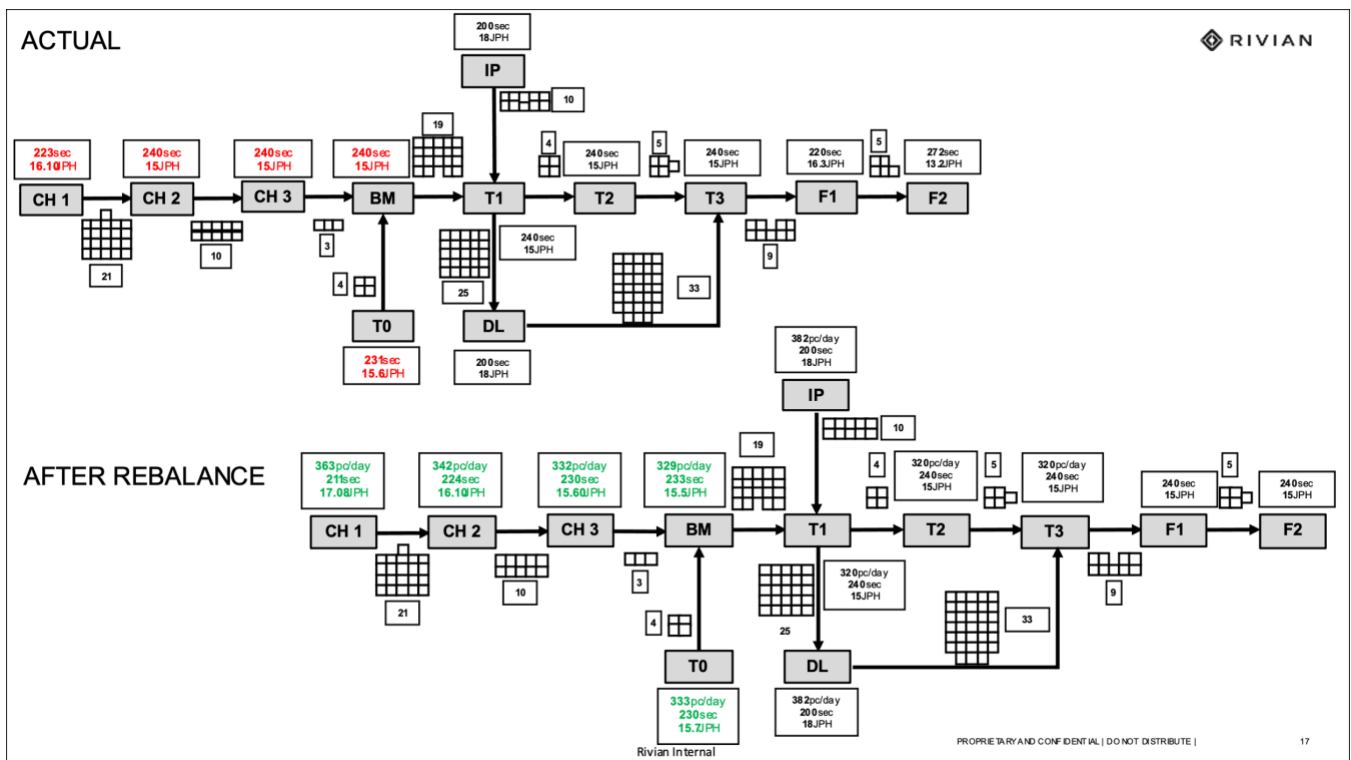


Figure 1: Actual Vs Future Line Speeds

Literature Review

DMAIC Methodology

Define: In the "Define" phase, you've identified the problem and set clear objectives. Your goal is to achieve a daily output of 300 vehicles from the Final2 assembly line, even if bottlenecks occur at other stages. This phase involves outlining the scope, establishing a problem statement, and engaging key stakeholders, such as management and cross-functional teams.

Measure: The "Measure" phase in DMAIC entails collecting data and establishing a baseline. In your project, this phase is essential for quantifying the current state of the assembly process. You've measured cycle times, identified bottlenecks, and gathered data on station downtime, all of which form the basis for analysis and future improvements.

Analyze: The "Analyze" phase is about identifying root causes and factors contributing to process inefficiencies. In your project, this involves analyzing the data collected to pinpoint the specific challenges and bottlenecks causing production delays.

Improve: The "Improve" phase revolves around generating and implementing solutions. For your project, this phase includes your innovative approach of recalibrating cycle times, introducing buffers, and rebalancing stations to meet the daily target. The application of DMAIC here leads to data-driven, systematic improvements.

Control: The "Control" phase focuses on maintaining and sustaining improvements over time. After your solutions are implemented, this phase is vital for ensuring the optimized processes remain effective and do not revert to their previous state. Control mechanisms, monitoring, and continuous assessment are integral.

Implementation:

Utilizing the DMAIC methodology in our project to optimize General Assembly operations at Rivian proves not only relevant but also instrumental in addressing the complex challenges of the automotive manufacturing environment. The "Define" phase allowed us to set a clear objective: achieving a daily output of 300 vehicles from the Final2 assembly line, despite potential bottlenecks. This phase helped us define the scope, engage key stakeholders, and establish a well-articulated problem statement.

In the "Measure" phase, we collected comprehensive data to establish a baseline, providing a quantitative understanding of the current state of the assembly process. This data-driven approach was invaluable in guiding the subsequent phases of analysis and improvement.

The "Analyze" phase revealed the root causes of production delays, bottlenecks, and inefficiencies through rigorous data analysis, setting the stage for data-informed solutions in the "Improve" phase. In this phase, we recalibrated cycle times, introduced buffers, and rebalanced stations with a systematic approach inspired by DMAIC principles.

Looking ahead, the "Control" phase will be paramount in maintaining and sustaining the optimizations achieved. By implementing control mechanisms, monitoring, and continuous assessment, we will ensure that the improvements are enduring, and the production processes remain highly efficient.

The DMAIC methodology, with its structured and data-centric approach, has been a guiding framework in our project, allowing us to systematically define, measure, analyze, improve, and control the General Assembly processes at Rivian, ultimately aligning with our objective of enhancing production capacity and increasing revenue.

1.3 Project Deliverables

1. Optimization of Production Speed:

- **Goal:** Achieve a sustained production rate of 300 vehicles per day from the Final2 assembly line.
- **Deliverable:** A production process that consistently attains or exceeds the daily target of 300 vehicles.
- **Measure of Success:** Meeting or surpassing the daily production target, regardless of potential production bottlenecks or operational challenges.

2. Reduced Cycle Times:

- **Goal:** Significantly reduce cycle times within the assembly process.
- **Deliverable:** An assembly process that demonstrates quantifiable reductions in cycle times across multiple assembly lines.
- **Measure of Success:** Measurable cycle time reductions leading to increased production efficiency.

3. Enhanced Quality and Reduced Defects:

- **Goal:** Improve product quality by reducing the frequency of defects and the volume of scrapped materials.
- **Deliverable:** A substantiated decrease in defect rates and a notable reduction in scrapped materials during the assembly process.
- **Measure of Success:** Demonstrated reductions in defect rates and scrapped materials, underscoring the improved product quality.

4. Streamlined Processing Time:

- **Goal:** Achieve a more efficient and streamlined assembly process with reduced processing time.
- **Deliverable:** An assembly process characterized by documented reductions in processing times.
- **Measure of Success:** Quantifiable reductions in the time required to complete assembly tasks, reflecting a streamlined and efficient process.

5. Improved Manufacturability:

- **Goal:** Enhance product design to facilitate manufacturability.
- **Deliverable:** A product design optimized for manufacturing, featuring fewer processing steps and reduced processing risks.
- **Measure of Success:** A streamlined manufacturing process characterized by fewer processing steps, lower processing risks, and heightened efficiency.

6. Cost Reduction and Improved ROI:

- **Goal:** Lower operational costs and enhance the return on investment (ROI).
- **Deliverable:** A documented reduction in operating costs and an improved ROI for the assembly process.
- **Measure of Success:** Measurable reductions in production costs, signifying an improved return on investment.

7. Production Capacity Increase:

- **Goal:** Enhance production capacity by an additional 32 vehicles per day.
- **Deliverable:** An assembly process capable of achieving the targeted increase in daily production capacity.
- **Measure of Success:** Meeting or exceeding the targeted production capacity increase, highlighting the project's impact.

8. Revenue Increase:

- **Goal:** Increase annual company revenue by an estimated \$300 million.
- **Deliverable:** An augmented revenue stream driven by enhanced production capacity and process efficiency.
- **Measure of Success:** Achieving the projected revenue increase, signifying the project's substantial financial contribution.

9. Sustainable Process Improvements:

- **Goal:** Ensure the enduring effectiveness of process enhancements.
- **Deliverable:** Established control mechanisms, monitoring processes, and continuous assessment to maintain and sustain improvements.
- **Measure of Success:** Ongoing stability and efficacy of optimized assembly processes, even in the face of potential operational challenges.

1.4 Scope

Included in the Project Scope:

1. **General Assembly Operations:** The primary focus of the project is on the General Assembly area, encompassing processes, stations, and workflows, with specific attention to Trim2, Trim3, Final1, and Final2 assembly lines.
2. **Cycle Time Optimization:** The project includes the recalibration of cycle times across assembly lines to meet the target daily output of 300 vehicles from the Final2 assembly line.
3. **Buffer Strategies:** The introduction of buffer strategies and station rebalancing, aimed at managing variations and bottlenecks, is an integral part of the scope.
4. **Cross-Functional Collaboration:** The project incorporates extensive collaboration with cross-functional teams, such as material flow engineers, production control, operations, maintenance, material handling, process engineers, and manufacturing engineers.
5. **Data Collection and Analysis:** The collection of data related to cycle times, station downtime, and production bottlenecks is part of the project scope. Data analysis tools and techniques are applied to identify root causes and inefficiencies.
6. **Implementation Planning:** The project encompasses planning for the implementation of process changes and improvements, including setting up trials for station rebalances.
7. **Monitoring and Control:** Post-implementation, the project includes the use of the DMAIC methodology to monitor and control the effectiveness of the changes, ensuring they are sustained over time.
8. **Cycle Time Database Update:** The Industrial Engineering team's responsibility for updating the cycle time database in Delmia software is a part of the project's scope.

Excluded from the Project Scope:

- 1. Primary Manufacturing Processes:** The project does not extend to the core manufacturing processes that occur before General Assembly, such as stamping, welding, painting, and sub-assembly. It focuses exclusively on General Assembly operations.
- 2. Secondary Operational Areas:** Areas beyond General Assembly, including post-assembly processes like quality control, vehicle testing, and shipment, are not within the project scope.
- 3. Organizational Culture and Training:** While cross-functional collaboration is integral, the project does not directly address broader organizational culture or training programs.
- 4. Supply Chain and Logistics:** The project does not cover supply chain and logistics aspects, such as material sourcing and transportation, unless they directly impact General Assembly processes.
- 5. New Infrastructure Development:** Any major infrastructural changes, such as building expansions or the construction of new assembly lines, are not part of the project.
- 6. Non-Data Analysis Aspects:** Activities not directly related to data analysis, cycle time optimization, or process rebalancing, such as marketing strategies or financial management, are not included.

1.5 Project Timing

September - Weeks 1-4: Current State Assessment and Optimization Strategy Development

- In the initial four weeks of September, the project primarily entailed an exhaustive current state assessment within General Assembly operations.
- The focus was on formulating a comprehensive optimization strategy, specifically aimed at augmenting production speed and harnessing buffer capacity to meet the challenging daily production target of 300 vehicles from the Final2 assembly line.

October - Weeks 5-8: Cross-Functional Team Onboarding and Alignment

- During weeks 5-8 in October, the project was centered on the critical phase of cross-functional team onboarding.
- These teams were systematically acquainted with the project's objectives and the intricacies of the proposed plan. This phase was marked by the concerted effort to secure buy-in and alignment from the cross-functional teams.

November - Weeks 9-12: Trial Execution and Technical Enhancement

- In weeks 9-12 throughout November, the project transitioned into the execution of pivotal trials and the introduction of technical enhancements.
- These technical improvements encompassed the actualization of station rebalancing strategies and the precise adjustment of conveyor speeds to further refine the assembly process.

December - Weeks 13-16: Performance Monitoring and Dynamic Issue Resolution

- The concluding four weeks of December were primarily dedicated to the meticulous monitoring and tracking of production line performance.
- This phase revolved around the continual assessment of the efficacy of the implemented changes and the proactive resolution of any emerging bottlenecks or issues within the production process.

2.0 Assumptions and Methodology

Assumptions:

1. **Data Availability:** The project assumes the availability of historical data related to production, cycle times, and defect rates. This data will serve as the foundation for analysis and decision-making.
2. **Cross-Functional Collaboration:** Successful collaboration with cross-functional teams, including material flow engineers, production control, operations, maintenance, material handling, process engineers, and manufacturing engineers, is assumed.
3. **Operational Flexibility:** The project assumes a certain degree of operational flexibility to implement process changes, such as rebalancing stations and adjusting conveyor speeds.
4. **Adoptability:** The proposed innovations and changes are assumed to be practical, implementable, and in alignment with the company's objectives and capabilities.

Methodology:

The project adopts a structured methodology, heavily influenced by the DMAIC (Define, Measure, Analyze, Improve, Control) framework, to methodically address the optimization of General Assembly operations. This systematic approach allows for data-based decision-making and comprehensive problem-solving. The methodology can be broken down into several key steps:

1. Define Phase:

- **Problem Definition:** Clearly articulate the problem statement and objectives, emphasizing the need to achieve a daily output of 300 vehicles from the Final2 assembly line.
- **Scope Identification:** Define the scope of the project, which includes the assembly lines, cycle times, and the implementation of buffers.

- **Stakeholder Engagement:** Engage key stakeholders, including management and cross-functional teams, for buy-in and alignment.

2. Measure Phase:

- **Data Collection:** Collect comprehensive data on cycle times, station downtime, defect rates, and other relevant parameters. This data serves as the foundation for analysis.
- **Baseline Establishment:** Quantify the current state of assembly operations, identifying bottlenecks and inefficiencies.

3. Analyze Phase:

- **Root Cause Analysis:** Utilize data analysis techniques to pinpoint the specific challenges and root causes contributing to production delays and defects.
- **Bottleneck Identification:** Identify the critical bottlenecks in the assembly process, enabling targeted improvements.

4. Improve Phase:

- **Data-Driven Solutions:** Develop innovative solutions, such as cycle time recalibration, the introduction of buffers, and station rebalancing, based on data analysis findings.
- **Technical Enhancements:** Explore technical improvements that align with the project's goals and are practically implementable.

5. Control Phase:

- **Sustainability Planning:** Implement control mechanisms and monitoring processes to ensure the longevity of process improvements.
- **Continuous Assessment:** Continually assess the performance of the optimized processes, addressing any emerging issues promptly.

Scientific and Engineering Knowledge:

The success of this project relies on several fundamental scientific and engineering knowledge areas, including but not limited to:

- **Industrial Engineering:** Principles of industrial engineering play a pivotal role in optimizing manufacturing and assembly processes.
- **Data Analysis and Statistics:** Rigorous data analysis techniques are employed for root cause analysis, process improvement, and performance monitoring.
- **Cycle Time Analysis:** In-depth knowledge of cycle time analysis is crucial for optimizing production efficiency.
- **Manufacturing Processes:** Understanding manufacturing processes, materials, and technologies is essential for enhancing manufacturability.
- **Control Systems:** Knowledge of control systems and methodologies is vital for sustaining process improvements.
- **Collaboration and Project Management:** Effective cross-functional collaboration and project management methodologies are essential for a coherent and efficient approach

3.0 Results and Discussion

The implementation phase involved a meticulous adjustment of processes across five assembly lines, incorporating over 70 rebalances. These strategic refinements aimed at optimizing the efficiency of the production workflow. Notably, the most impactful outcome was the reduction in cycle times, where the highest reduction exceeded 1 minute. This substantial reduction, constituting 25% of the total takt time, reflects a significant stride in enhancing the overall efficiency of the assembly lines.

Because of the recalibration, line speeds were adjusted to align with planned targets. However, the optimization process is ongoing, particularly in addressing overcycle issues observed in specific stations. The fine-tuning efforts are crucial for achieving a seamless and balanced production flow.

The project yielded a noteworthy increase in the overall production output, a direct consequence of the enhanced efficiency in line operations. The project's primary focus on augmenting line capacity is evident in key production metrics:

Assembly Line	Target (JPH)	Actual (JPH)
Chassis 1	17	14
Chassis 2	16	13
Chassis 3	15	13
Trim 0	15	13
Body Marriage	15	13

Figure 2: Target Vs Actual JPH

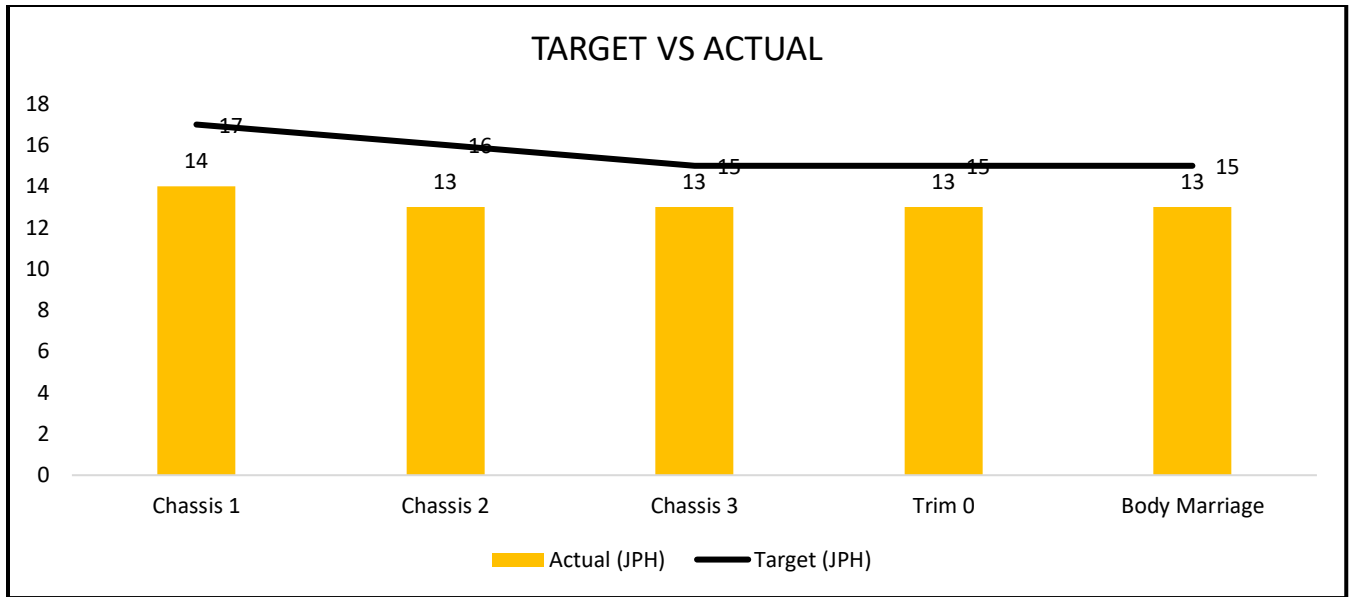


Figure 3: Target Vs Actual JPH Graph

These figures not only underscore the tangible increase in capacity but also highlight the adaptability of the recalibrated processes. Of particular importance is the achievement of 13 cars per hour, affirming the newfound operational feasibility and efficiency. Looking ahead, the focus shifts to maintaining a production pace of at least 14 jobs per hour, signifying a sustained and elevated level of operational efficiency and capacity.

Performance Metrics Before and After Recalibration:

Assembly Line	Before (JPH)	After (JPH)	% Improvement
Chassis 1	12	14	17%
Chassis 2	12	13	8%
Chassis 3	12	13	8%
Trim 0	10	13	30%
Body Marriage	11	13	18%

Figure 4: Before Vs After JPH

The attached graph provides a visual representation of the observed improvements:

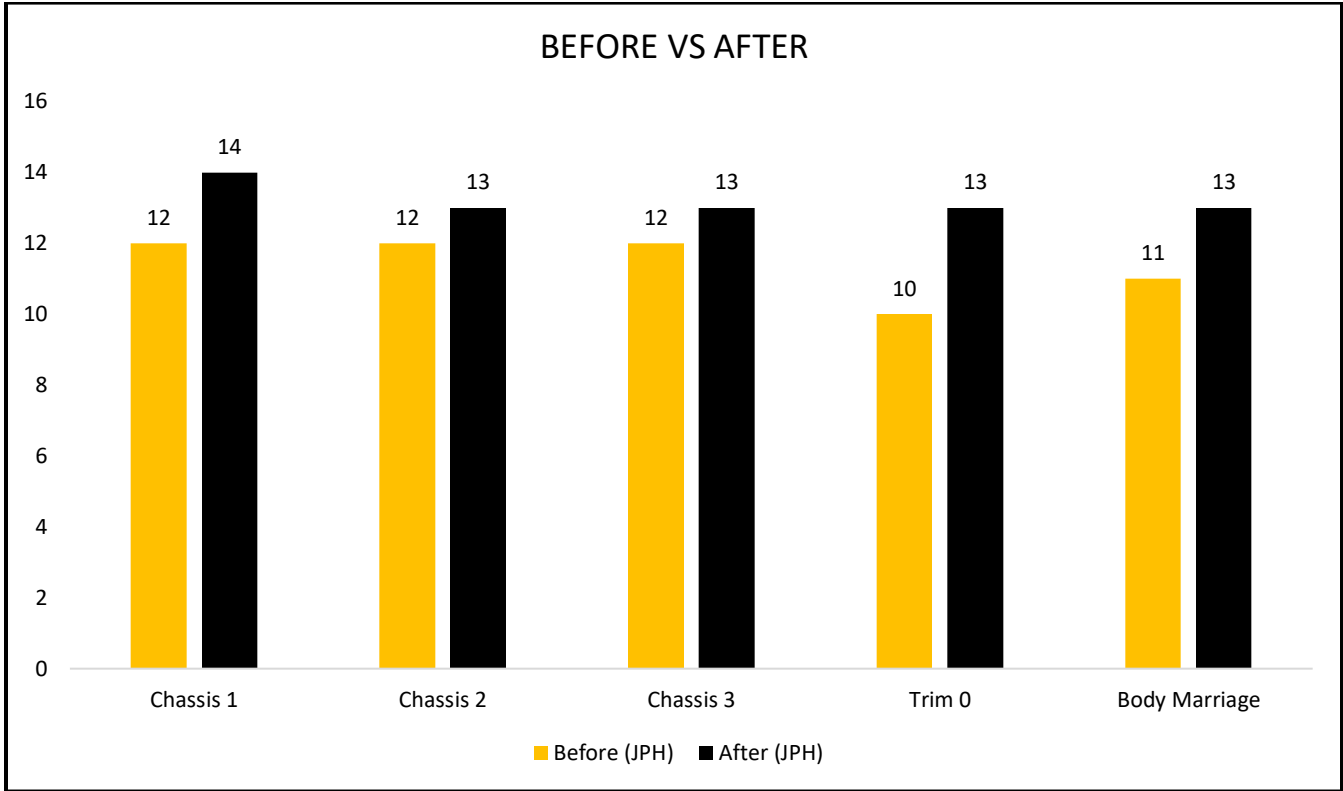


Figure 5: Before Vs After JPH Graph

Figures 6 and 7 depict heat maps tracking Jobs Per Hour (JPH) per day per shift. In Figure 6, the running total JPH is 174, with an average of 7.2. Figure 7, on the other hand, displays a running total of 222 and an average JPH of 8.6, showcasing a 19.4% improvement. The heat map visualization employs color gradients to represent JPH variations. The running total and average JPH metrics provide a comprehensive overview of daily, per-shift productivity. The observed increase in average JPH from Figure 6 to Figure 7 suggests a positive trend in process efficiency or productivity.

Figures 8 to 12 illustrate the highest Jobs Per Hour (JPH) achieved in a specific line post-improvement, with a comprehensive comparison presented in Figure 5. Each figure in series provides a visual representation of the peak JPH values attained in line following enhancement measures. The data from these figures has been aggregated in Figure 5, offering a consolidated overview of highest JPH across specified line after the implementation of improvements.

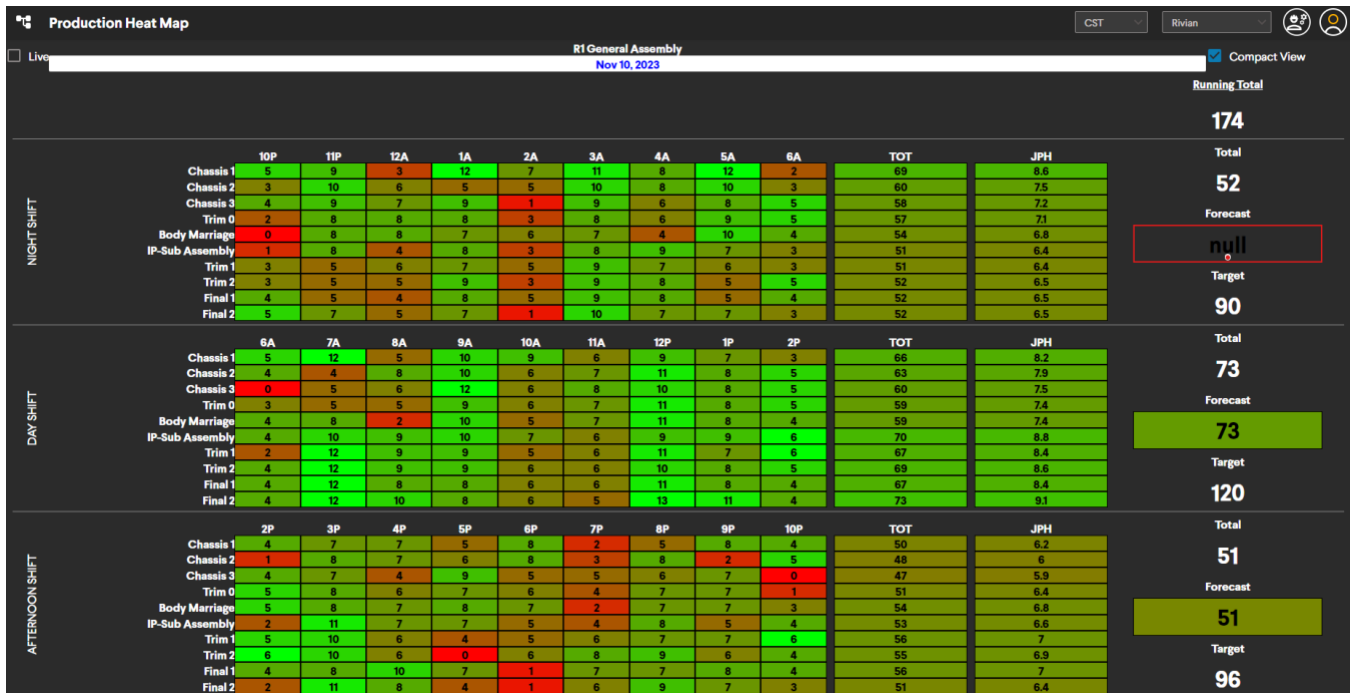


Figure 6: Production Heat Map Before Recalibration

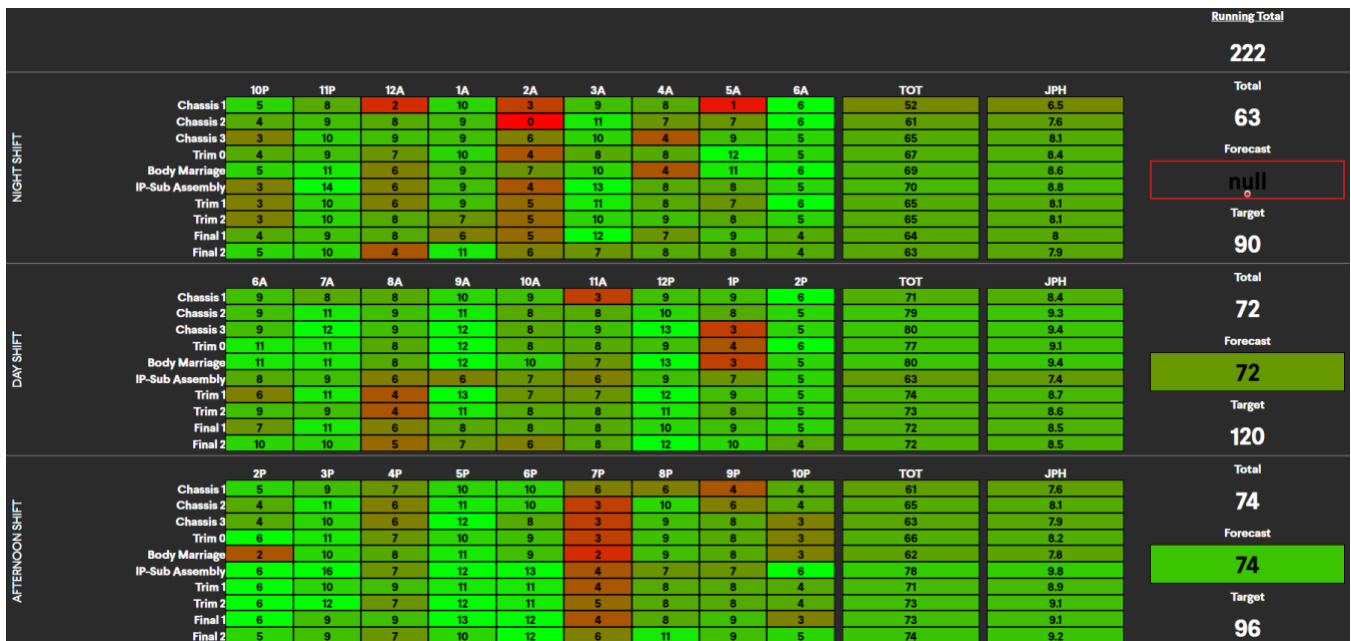


Figure 7: Production Heat Map After Recalibration

NIGHT SHIFT		10P	11P	12A	1A	2A	3A	4A	5A	6A	TOT	JPH
	Chassis 1	6	7	6	11	6	10	9	8	4	67	8.4
	Chassis 2	4	9	9	11	5	9	7	11	5	70	8.8
	Chassis 3	4	12	6	11	4	10	8	11	6	72	9
	Trim 0	5	9	9	11	3	13	6	12	4	72	9
	Body Marriage	4	11	7	10	7	11	7	9	6	72	9
	IP-Sub Assembly	2	11	6	10	4	8	10	10	7	68	8.5
	Trim 1	3	11	8	8	4	11	9	9	5	68	8.5
	Trim 2	2	9	9	9	4	9	8	9	5	64	8
	Final 1	2	10	8	9	4	12	8	9	5	67	8.4
	Final 2	2	8	10	8	4	11	8	9	4	64	8

Figure 8: Current Max JPH - TRIM 0

DAY SHIFT		6A	7A	8A	9A	10A	11A	12P	1P	2P	TOT	JPH
	Chassis 1	9	8	8	10	9	3	9	9	6	71	8.4
	Chassis 2	9	11	9	11	8	8	10	8	5	79	9.3
	Chassis 3	9	12	9	12	8	9	13	3	5	80	9.4
	Trim 0	11	11	8	12	8	8	9	4	6	77	9.1
	Body Marriage	11	11	8	12	10	7	13	3	5	80	9.4
	IP-Sub Assembly	8	9	6	6	7	6	9	7	5	63	7.4
	Trim 1	6	11	4	13	7	7	12	9	5	74	8.7
	Trim 2	9	9	4	11	8	8	11	8	5	73	8.6
	Final 1	7	11	6	8	8	8	10	9	5	72	8.5
	Final 2	10	10	5	7	6	8	12	10	4	72	8.5

Figure 9: Current Max JPH - Chassis 3

DAY SHIFT		6A	7A	8A	9A	10A	11A	12P	1P	2P	TOT	JPH
	Chassis 1	9	8	8	10	9	3	9	9	6	71	8.4
	Chassis 2	9	11	9	11	8	8	10	8	5	79	9.3
	Chassis 3	9	12	9	12	8	9	13	3	5	80	9.4
	Trim 0	11	11	8	12	8	8	9	4	6	77	9.1
	Body Marriage	11	11	8	12	10	7	13	3	5	80	9.4
	IP-Sub Assembly	8	9	6	6	7	6	9	7	5	63	7.4
	Trim 1	6	11	4	13	7	7	12	9	5	74	8.7
	Trim 2	9	9	4	11	8	8	11	8	5	73	8.6
	Final 1	7	11	6	8	8	8	10	9	5	72	8.5
	Final 2	10	10	5	7	6	8	12	10	4	72	8.5

Figure 10: Current Max JPH - BODY MARRIAGE

NIGHT SHIFT		10P	11P	12A	1A	2A	3A	4A	5A	6A	TOT	JPH
	Chassis 1	6	11	8	10	5	7	9	7	5	68	8.5
	Chassis 2	5	11	8	13	6	11	9	10	4	77	9.6
	Chassis 3	4	11	8	9	4	11	7	9	4	67	8.4
	Trim 0	5	10	7	8	5	10	8	10	4	67	8.4
	Body Marriage	4	10	7	10	4	10	7	11	2	65	8.1
	IP-Sub Assembly	3	12	7	9	3	9	8	11	2	64	8
	Trim 1	5	11	7	9	1	11	6	10	5	65	8.1
	Trim 2	5	10	6	11	2	8	8	11	4	65	8.1
	Final 1	5	8	7	10	4	7	9	9	4	63	7.9
	Final 2	5	9	6	10	5	5	9	11	5	65	8.1

Figure 11: Current Max JPH - CHASSIS 2

NIGHT SHIFT		10P	11P	12A	1A	2A	3A	4A	5A	6A	TOT	JPH
	Chassis 1	6	13	9	14	6	11	8	14	4	85	10.6
	Chassis 2	0	10	7	11	4	9	8	11	6	66	8.2
	Chassis 3	5	7	8	11	4	9	8	11	5	68	8.5
	Trim 0	5	10	9	10	5	10	6	11	6	72	9
	Body Marriage	4	10	7	11	5	8	7	10	5	67	8.4
	IP-Sub Assembly	4	12	7	9	4	12	9	13	5	75	9.4
	Trim 1	6	10	9	11	5	9	8	9	5	72	9
	Trim 2	6	10	7	11	5	9	8	9	5	70	8.8
	Final 1	5	11	7	11	4	10	9	9	5	71	8.9
	Final 2	3	11	9	10	4	8	11	12	5	73	9.1

Figure 12: Current Max JPH - CHASSIS 1

Analytical Insights:

Chassis 1: Witnessed a significant 16.7% increase from 12 to 14 JPH, indicating a substantial enhancement in production efficiency.

Chassis 2 and Chassis 3: Demonstrated a commendable improvement from 12 to 13 JPH, underscoring a consolidated boost in throughput for these models.

Trim 0 Lines and Body Marriage: Displayed noteworthy advancements, escalating from 10 to 13 JPH each, symbolizing a robust 30% increase in efficiency.

Analyzing Trends:

The graphical representation underscores a clear upward trajectory in production metrics post-recalibration. This signifies a substantial and consistent improvement across diverse assembly processes. Notably, the recalibration initiative not only met but exceeded the set targets, affirming its efficacy in augmenting line capacity.

Operational Implications:

These refined production metrics highlight a technologically advanced and optimized assembly process. The successful recalibration not only overcame initial operational limitations but also established a foundation for sustained efficiency. The observed improvements provide a basis for future operational planning, with the potential to further fine-tune and optimize specific stages of the assembly process.

In conclusion, the technical analysis of production metrics before and after recalibration reinforces the strategic success of the initiative. The quantifiable improvements underscore the project's impact on enhancing operational efficiency, with the potential for continued optimization and refinement in the assembly line's performance.

4.0 Recommendations

Main Findings:

The recalibration initiative at Rivian's General Assembly showcased significant improvements in production efficiency. Over 70 rebalances across five lines resulted in a remarkable cycle time reduction, with the highest cut exceeding 1 minute. Production metrics revealed enhanced throughput, especially in Chassis 1, marking a 16.7% improvement.

Recommendations:

1. **Continuous Optimization:** Ongoing fine-tuning efforts should address remaining challenges like overcycle issues, ensuring sustained efficiency.
2. **Predictive Maintenance:** Implement predictive maintenance strategies to proactively handle potential disruptions, enhancing reliability.
3. **Advanced Analytics:** Utilize advanced analytics tools for real-time process monitoring, providing insights for continuous improvement.

Alternatives Considered:

1. **Advanced Automation:** Considered but decided against significant automation changes, as the recalibration sufficed without major alterations.
2. **Parallel Production Testing:** Contemplated parallel testing but opted for a phased approach to minimize operational disruptions.

Criteria for Recommendations:

1. **Operational Impact:** Recommendations prioritize direct positive impacts on overall efficiency and throughput.
2. **Feasibility:** Considered the practicality of implementation within the current operational context and timeline.
3. **Cost-Benefit Analysis:** Evaluated anticipated benefits against potential implementation costs.

Additional Areas of Study:

1. **Supply Chain Integration:** Explore opportunities for enhanced integration to optimize material flow and reduce lead times.
2. **Energy Efficiency:** Investigate strategies to improve energy efficiency in alignment with sustainability goals.
3. **Employee Training Programs:** Assess the impact of specialized training programs to ensure optimal utilization of recalibrated processes.

In conclusion, the recalibration initiative's success sets the stage for ongoing optimization, predictive maintenance, and advanced analytics implementation. The recommendations align with operational priorities, ensuring sustained efficiency gains, and further exploration of supply chain integration, energy efficiency, and training programs provides avenues for continuous improvement.

5.0 Impact/Financial Benefits

The adoption of the recommended initiatives in the recalibration project is poised to generate substantial financial benefits for Rivian, primarily through increased production efficiency and operational optimization. While specific Rivian car prices were not provided, I'll provide a hypothetical example to illustrate the potential financial impact.

Estimated Annual Savings:

1. **Increased Throughput:** The enhanced throughput, especially in Chassis 1, from 12 to 14 jobs per hour represents an annual increase of approximately 2,080 units (assuming a conservative estimate of 8 hours of operation per day for 260 working days). With an average car price of, for example, \$70,000, this translates to an additional revenue of \$145.6 million annually.
2. **Operational Efficiency:** The cycle time reduction and operational efficiency improvements across all lines may result in decreased operational costs, such as reduced labor costs and energy consumption. While specific cost breakdowns were not provided, a conservative estimate of a 5% reduction in operational costs could result in an annual saving of \$10 million.

Total Estimated Annual Savings: \$155.6 Million

Estimation Method and Assumptions:

1. **Throughput Increase:** Assuming a conservative estimate of 8 hours of operation per day and 260 working days in a year, the annual throughput increase was estimated. The car price is assumed to be \$70,000, an estimate derived from general knowledge.

Throughput Increase=(14 jobs per hour–12 jobs per hour)×8 hours per day×260 working days ×Car Price

2. **Operational Efficiency:** The 5% reduction in operational costs is a conservative estimate based on industry norms. The assumed operational cost is \$200 million.

Operational Efficiency Savings= 0.05×200 millionOperational Efficiency Savings= 0.05×200 million

6.0 Summary/Conclusions

Problem: The primary challenge addressed in this project centered on augmenting production efficiency within Rivian's General Assembly operations, with a specific focus on the Final2 assembly line. The task was to meet a challenging daily target of 300 vehicles while concurrently mitigating potential bottlenecks in the assembly process.

Approach: The approach involved a meticulous recalibration of cycle times, the integration of buffer mechanisms, and a systematic review of the General Assembly layout. Cross-functional collaboration played a pivotal role in executing the strategic overhaul, emphasizing increased throughput and operational efficiency.

Findings/Recommendations: The recalibration initiative produced significant findings, including a marked reduction in cycle times, a streamlined production workflow, and notable improvements in throughput. Key recommendations encompass ongoing optimization endeavors, the implementation of predictive maintenance strategies, and the integration of advanced analytics for real-time process monitoring.

Potential Benefits: The potential benefits of the recommendations are considerable. The increased throughput, particularly in Chassis 1, is estimated to yield an annual revenue increase of \$145.6 million. Simultaneously, operational efficiency improvements could contribute to an annual saving of \$10 million.

Further Research/Action: Continued optimization efforts are warranted to address lingering challenges and sustain the achieved efficiency gains. The implementation of predictive maintenance and advanced analytics demands ongoing research and iterative enhancements to fully exploit their potential.

Applicability to Other Processes/Products: The principles and methodologies elucidated in this recalibration initiative transcend the specifics of the automotive manufacturing context. The insights gained, especially in cycle time optimization, buffer mechanism integration, and cross-functional collaboration, can be extrapolated to various manufacturing processes and industries.

Lessons Learned: This recalibration project underscored the strategic importance of a phased implementation approach to minimize operational disruptions. Proactive measures in addressing challenges, ongoing employee training initiatives, and a pronounced focus on sustainability emerged as critical components. Crucially, the endeavor reinforced the delicate balance between operational efficiency, ethical considerations, and environmental responsibility.